

III. EDA-Data Cleaning

AIM:

- Handling missing values: detection, filling, and dropping
- Removing duplicates and unnecessary data
- Data type conversion and ensuring consistency
- Normalize data (e.g., standardization, min-max scaling).

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler

# Load the dataset
df = pd.read_csv('/content/test_Y3wMUE5_7gLdaTN.csv')

# Display basic information
print("Initial Data Overview:")
print(df.info())

# 1. Handling Missing Values
print("\nMissing Values in Each Column:\n", df.isnull().sum())

plt.figure(figsize=(10, 4))
sns.heatmap(df.isnull(), cbar=False, cmap="Blues")
plt.title("Missing Value Heatmap")
plt.show()
```

Initial Data Overview:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 367 entries, 0 to 366

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	Loan_ID	367 non-null	object
1	Gender	332 non-null	object
2	Married	367 non-null	object
3	Dependents	356 non-null	object
4	Education	367 non-null	object
5	Self_Employed	335 non-null	object
6	ApplicantIncome	367 non-null	int64
7	CoapplicantIncome	367 non-null	int64
8	LoanAmount	344 non-null	float64
9	Loan_Amount_Term	346 non-null	float64
10	Credit_History	336 non-null	float64
11	Property_Area	367 non-null	object

dtypes: float64(3), int64(2), object(7)

memory usage: 34.5+ KB

None

Missing Values in Each Column:

Loan_ID	0
Gender	35
Married	0
Dependents	11
Education	0
Self_Employed	32
ApplicantIncome	0
CoapplicantIncome	0
LoanAmount	23
Loan_Amount_Term	21
Credit_History	31

```
dtype: int64
```

